

Speed of sound and temperature in the ocean by Brillouin scattering

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A method is described to measure the speed of sound and the temperature in the sea as functions of depth. Backscattered laser light is analyzed with an interferometric spectrometer. The speed of sound at very short acoustic wavelengths is obtained directly from the wavelength shift of the Brillouin scattered light, and the temperature is deduced from the speed of sound together with auxiliary information on depth and salinity. Experiments are described.

I. Introduction

There are a number of reasons why it is important to measure subsurface temperature^{1,2} and speed of sound, but there are no techniques routinely in use to make these measurements accurately, cheaply, and rapidly. Ship soundings, while accurate, are very time-consuming, and expendable bathythermographs are faster but expensive. A rapid remote-sensing method would be very desirable.

The Raman effect³⁻¹⁰ shows promise, but its spectrum is very broad, which may be a disadvantage when interferences such as fluorescence, daylight, or selective absorption (e.g., gelbstoff) are present.

The Brillouin spectrum, characterized by very sharp lines, has also been suggested.¹¹⁻¹⁷ The narrow lines tend to be free from interference caused by fluorescence, daylight, or selective absorption. To take advantage of these properties, however, observations must be made with interferometric precision. This paper describes the methods developed.

II. Rayleigh Scattering

Phenomena comprising Rayleigh scattering are characterized by very small wavelength changes (<0.1 nm), and its characteristics have been calculated by a number of authors,¹⁸⁻²⁰ so only a summary will be given here. Of the possible phenomena which exist, only a few play a significant role in liquid water. These in-

clude isobaric, Brillouin, and particulate scattering. For standard conditions (salinity 35 ppt, 20°C, and 1 atm) isobaric scattering gives rise to a central (unshifted) peak with the volume scattering coefficient,¹⁹ $\beta(180)_i = 1.67 \times 10^{-5} \text{ m}^{-1} \text{ sr}^{-1}$ and breadth, $\Gamma_r = 1.7 \times 10^{-3} \text{ cm}^{-1}$ at 20°C.

Pressure waves give rise to the two Brillouin²¹ peaks with a separation

$$\Delta\nu_B = (2n\nu_s/c)\nu_0 \sin(\Theta/2) \text{ cm}^{-1}, \quad (2)$$

where n is the index of refraction of the liquid, ν_s is the speed of sound, ν_0 is the wave number of the incident radiation, and Θ is the scattering angle. For standard conditions the breadth of the backscattered Brillouin components Γ_B is $1.6 \times 10^{-2} \text{ cm}^{-1}$, while $\Delta\nu_B$ is 0.27 cm^{-1} . The volume backscattering coefficient for the Brillouin components, $\beta(180)_B$, is given by^{11,19}

$$\beta(180)_B = 2.4 \times 10^{-4} \text{ m}^{-1} \text{ sr}^{-1}. \quad (3)$$

Although the total signal strengths from Raman and Brillouin backscattering are nearly the same, efficient narrowband detection of the Brillouin scattering has the potential for improving the signal-to-noise ratio, since the intensity per unit wavelength interval is much greater.

For measuring depths to within a resolution of 1 m, laser pulse widths of ~ 10 nsec are necessary. The resultant interruption broadening is ~ 0.3 of the Brillouin breadth Γ_B . Assuming quasi-Gaussian shapes, this will broaden the resulting line by $\sim 4\%$, which can easily be corrected for since the shift $\Delta\nu_B$ is 16 times larger.

Particulate matter gives rise to a practically un-broadened central component which, except in very clean water, is stronger than the isobaric part of the Rayleigh scattering. Its intensity varies with the amount of particulate matter and can be used as a measure of the volume scattering coefficient for par-

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ticulates, after the small known isobaric fraction has been subtracted.

Other possible sources of scattering, including shear waves, thermal relaxation, anisotropy fluctuations, and salinity fluctuations, give rise to very small components which can be ignored and are not considered further.

The scattered beam, therefore, consists of a central line, flanked by two side components. The speed of sound may be deduced from the wavelength separation of the shifted components, while turbidity can be essentially obtained from the intensity ratio of the central to the side components.

Detailed calculations using a single scattering model are reported in Ref. 11. Conclusions of interest for remote sensing are that the Brillouin peaks are strong (>3 orders of magnitude more intense per wavelength interval than, for example, the Raman band), sharp (width less than a tenth of separation), and therefore are not appreciably influenced by external sources (sunlight, fluorescence) or transmission slopes. The spectrum in Fig. 1 was made using a Fabry-Perot interferometer and shows the Brillouin components for a sample of ordinary tap water.

III. Temperature Determination

The speed of sound in ocean water depends on the salinity, the pressure, and the temperature. A simplified equation,²³ plotted in Fig. 2, is

$$V_s = 1449 + 4.6T - 0.055T^2 + 0.003T^3 \\ + (S - 35)(1.39 - 0.012T) + 0.01z + \dots,$$

where T is temperature in degrees Celsius, S is salinity in parts per thousands, z is depth in meters, and V_s is sound speed in meters/second. If the depth and salinity are known the temperature can be deduced from the speed of sound as measured from the Brillouin peak separation.

The depth may be measured by taking account of the elapsed time between an outgoing laser pulse and the returning backscattered light. The salinity problem is more complicated. In freshwater applications, such as rivers and lakes, $S = 0$. In the well-mixed ocean, S is very nearly constant, while in brackish coastal waters and when overlying pools of freshwater are present S must be measured for each case. In the following treatment, we take the accepted³⁰ value for S of $35 \pm \frac{1}{2}$ parts/thousand. This uncertainty will not interfere significantly with temperature calculations to $\pm \frac{1}{2}^\circ\text{C}$.

IV. Experimental Procedures

The site chosen for a field test was a dock adjacent to the Miami, Florida, harbor channel ~ 2.25 km shoreward of the ocean end. At this point the channel was ~ 450 m wide, and the dredged passage was 9.5 m deep and 375 m wide. At the dock, the depth was ~ 8 m at high water. Strong tidal currents exist, and flow rates of 1.5 m/sec were usual except for short periods at the turn of the tide. The water was very dirty, with moving bits of weed, plankton, and other material.

The light source used for the first experiments was an argon-ion laser producing 0.1 W at 488 nm. Brillouin fringes were first observed in May 1977¹³ using a pressure-swept Fabry-Perot interferometer. Freon was used to sweep the Fabry-Perot, and a photomultiplier was employed to measure the light. The data were measured at a depth of 2 m.

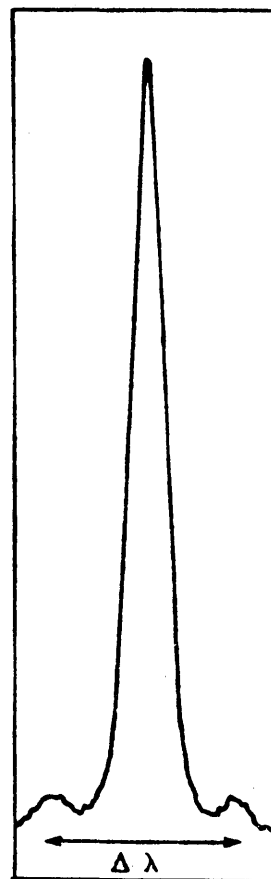


Fig. 1. Scattering in tap water showing Brillouin, Rayleigh, and Tyndall components. $\Delta\lambda \cong 1/5\lambda$. The spectrum was measured with a Fabry-Perot interferometer with a 1.00-cm spacer using light at 6328 Å from a He-Ne laser at a scattering angle of 135° .

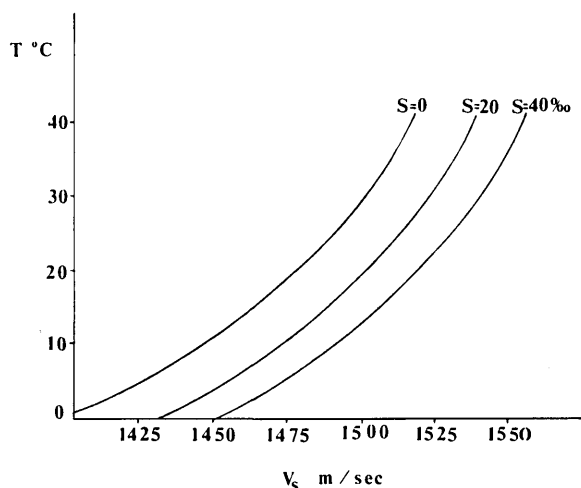


Fig. 2. Speed of sound at zero depth at various salinities for water in meters/sec is shown as a function of temperature (redrawn from Ref. 23).

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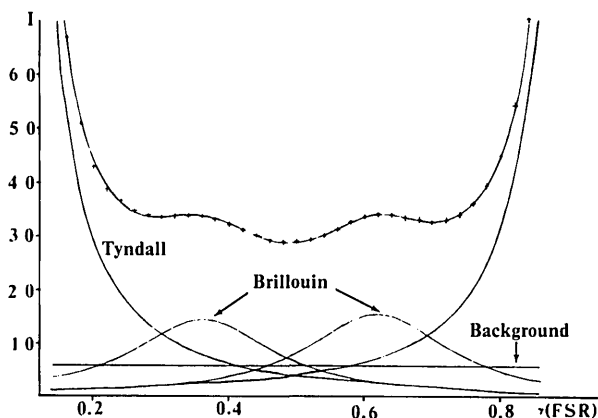


Fig. 3. Observations from natural seawater (averaged over seventeen Fabry-Perot orders) are fitted to theory. The crosses are observations, and the solid line is calculated.

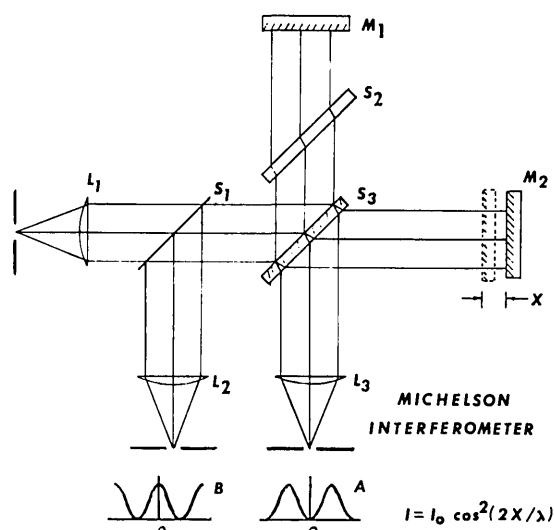


Fig. 4. Michelson interferometer showing collimating lens L_1 , 92% transmitting beam splitter S_1 , 50–50% beam splitter S_3 , compensating plate S_2 , mirrors M_1 and M_2 , with path difference x , and focusing lenses L_2 and L_3 . The interferometer is adjusted so that beam A (passing through L_3) contains the Brillouin components, while beam B (containing the central component). Since in ocean water the central component is strong, beam splitter S_1 needs to reflect only a small percentage of the light, saving most of it for beam A going to the Fabry-Perot interferometers.

To increase the signal-to-noise ratio, the data from seventeen selected orders were averaged. Experimental values at fifty optical path differences (shown as crosses in Fig. 3) were then compared to theoretical curves from the scattering theory. The computed curve using the best values for the Brillouin separation, instrumental finesse, and central peak strength (solid line in the graph) fits the experimental points as shown and the deduced temperature is $34 \pm 4^\circ\text{C}$ at an ambient temperature of 29°C as measured by a thermistor.

Although the field test showed that the Brillouin method could be used, it was necessary to improve the accuracy. A preliminary goal of $1/2^\circ\text{C}$ was chosen to demonstrate the practicability of this method.

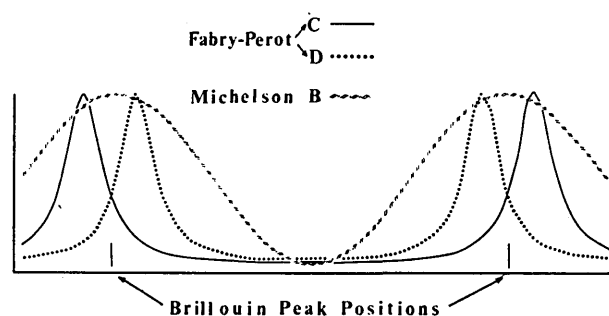


Fig. 5. This shows calculated passbands of the two Fabry-Perot interferometers together with that of the Michelson, as they are adjusted for channel A of Fig. 6. The central peak has been eliminated. As the two Brillouin peaks move further apart (higher speed of sound) the channel C intensity increases and that of channel D decreases. As they move closer together the reverse occurs. The information is contained in the ratio C/D , which is thus proportional to the speed of sound.

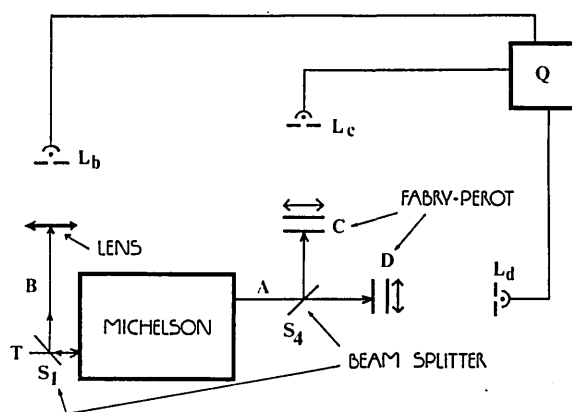


Fig. 6. Optical-electrical arrangement. Scattered light from the sea enters at T. The Michelson produces two beams as shown in Fig. 4. Backreflected beam B (containing the central peak (turbidity information) is partially reflected by S_1 , and falls on photomultiplier L_b producing signal B. Beam A contains the Brillouin peaks and is separated into two equal parts by 50–50% beam splitter S_4 . The light then passes to the two Fabry-Perots C and D. Q is a signal processor which provides the ratios C/D proportional to the speed of sound and $B/(C + D)$ proportional to turbidity.

A second experiment in the laboratory used a 6-W argon-ion laser with a Fabry-Perot mode-selecting stabilizer. A Michelson interferometer was added to reduce the effect of Tyndall scattering and make it possible to work in more turbid water. The Michelson interferometer separated the Brillouin peaks (beam A) from the unshifted Rayleigh-Tyndall light (beam B) by taking advantage of the zeros of the $(\cos)^2$ function (see Fig. 4). The situation for beam A is shown in detail in Fig. 5. The maxima of the Michelson $(\cos)^2$ function are so broad that they have a negligible effect on the shape of the much sharper Fabry-Perot peaks. The backreflected beam B, containing the unshifted Rayleigh-Tyndall component, is partially reflected by beam splitter S_1 (Fig. 6), consisting of a piece of flat glass (reflection coefficient $\sim 8\%$), into a photomultiplier L_b .

Table I. Characteristics of the Two Fabry-Perots Shown in Fig 6

Channel	Plate separation (cm)	Free spectral range (cm ⁻¹)
C	1.066	0.4967
D	0.824	0.6068

Beam A, containing only the Brillouin peaks, is directed to an equally dividing beam splitter, S_4 in Fig. 6, which illuminates the two Fabry-Perot interferometers, C and D. These are equipped with spacers fixing their free spectral ranges, as shown in Fig. 5 and Table I, with that for Fabry-Perot C slightly smaller and that for Fabry-Perot D slightly larger than twice the average Brillouin shift expected in the sea. If the wavelength shift increases, for example, in channel C the intensity rises and in D the intensity falls.

If now the ratio of the output of photomultipliers L_c and L_d is calculated, it will be nearly linearly proportional to the separation of the Brillouin components and therefore to the speed of sound. The ratio of the output of L_b to the sum of L_c and L_d , $B/(C + D)$ will in turn be proportional to the strength of the undeviated central component, which contains the information on particulates.

V. Experimental Results

The Fabry-Perots were adjusted as shown in Table I. Helium pressure was used to adjust the interferometers.

The temperature of the water bath was changed and measured with a Bureau of Standards traceable thermometer and the ratio of the channels C and D recorded with a time constant of 10 sec. Over 500 points were measured and typical results are shown in Fig. 7. The slope of the line in Fig. 7 is slightly more than 2% per degree. The jitter in the ratio L_c/L_d was $\sim 1\%$, allowing the temperature to be determined to $\pm 1/2^\circ\text{C}$.

VI. Conclusion

We have described and demonstrated in a preliminary series of experiments a method using Brillouin scattering for measuring the speed of short-wavelength sound in the ocean, which we believe will prove valuable. Because of the narrow spectral interval involved, some of the interferences that may affect the accuracy of the measurements are avoided.

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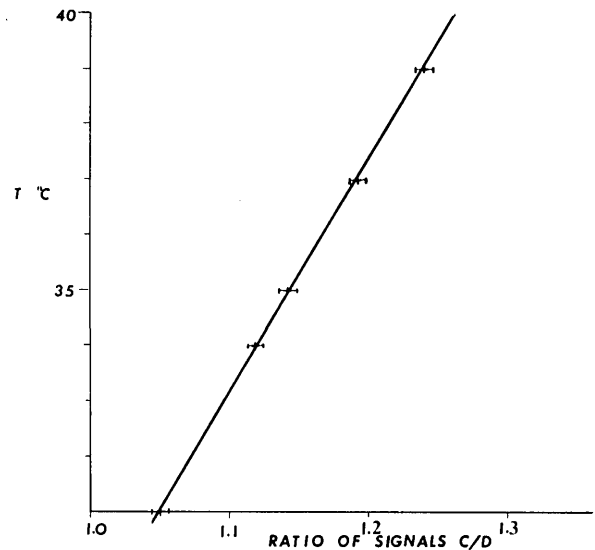


Fig. 7. Typical measurement made in the laboratory with the double Fabry-Perot method. Temperature in degrees Celsius, as measured with a standardized thermometer, is plotted against C/D , the ratio of the two Fabry-Perot photomultiplier signals.

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